Discount Signals vs. Satisfaction: Multimodal Prediction of Ratings and Sentiment on Amazon Product Listings

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CAP 5638 — Pattern Recognition — Spring 2026 — Florida State University

Abstract. This project investigates whether numeric discount signals (percentage, absolute markdown, price ratio) combined with review text can predict customer satisfaction on Amazon product listings. We address three research questions: (1) predicting star ratings via regression, (2) classifying review sentiment with and without pricing features, and (3) discovering product segments with heterogeneous discount–satisfaction relationships. Using a 1,464-row Amazon Sales Dataset from Kaggle, we build a multimodal pipeline combining TF-IDF text features with engineered discount facets via ColumnTransformer. An ablation study demonstrates that combined features outperform text-only (MAE 0.157 vs. 0.169) and discount-only variants for both regression and classification tasks. Linear models (Ridge, balanced SVC) outperform tree ensembles (XGBoost, HistGB, RandomForest) on this small, sparse-feature dataset. Unsupervised k-Means clustering (k=8) reveals product-type segments - including Smartphones. Unsupervised k-Means clustering (k=8) further yields interpretable soft segments with significantly different discount–satisfaction tendencies (ANOVA $p<10^{-8}$), linking behavioral pricing intuition with empirical pattern-recognition analysis.

1. Introduction

"X% OFF" banners flood e-commerce platforms, triggering fear-of-missing-out (FOMO) and impulse purchases. Yet deep discounts do not always produce satisfied customers as they can signal poor quality and erode brand equity. This project asks whether the combination of pricing cues and review language can predict customer satisfaction, and whether the answer varies by product segment.

The question has practical significance for merchandisers seeking data-driven pricing strategies. A small business owner who understands which product segments respond well to markdowns (and which do not) can avoid margin erosion while maintaining customer satisfaction. Imagine being able to estimate satisfaction from pricing data alone, before a single review is written that would enable proactive pricing adjustments rather than reactive damage control.

We formulate three research questions:

RQ1 (Rating Regression): Can we predict the numeric star rating (1–5) from discount signals, product metadata, and review text?

RQ2 (Sentiment Prediction): Can we classify review sentiment (positive vs. negative), and does adding pricing context improve accuracy?

RQ3 (Segment Heterogeneity): Does the relationship between discount depth and predicted rating vary across product categories or text-derived clusters?

2. Literature Review

Behavioral pricing research shows that consumers evaluate discounts relative to an internal reference benchmark rather than absolute price alone. Thaler's reference-price framework suggests that the perceived attractiveness of a deal depends on whether the markdown appears meaningfully favorable against expected value [1]. Related consumer research further finds that unusually deep discounts can alter perceptions of product quality, sometimes creating skepticism rather than satisfaction [2].

In modern digital retail environments, these pricing cues interact with product presentation, customer expectations, and online decision heuristics to shape purchasing behavior [3].

Text-based product rating prediction has also been widely studied in recommender systems and opinion mining. McAuley and Leskovec [4] show that review text contains latent information that improves the prediction of user ratings beyond structured metadata alone. More broadly, sentiment-analysis research demonstrates that customer language can be used to extract reliable satisfaction signals from online reviews [5]. These findings suggest that textual opinion and structured listing-side attributes may provide complementary information when modeling customer satisfaction.

Marketing research additionally indicates that discount-sensitive customers behave differently from quality-driven buyers, implying that markdown effectiveness is unlikely to be uniform across all product types [6]. This motivates examining not only whether discount variables improve predictive performance, but also whether the discount–satisfaction relationship varies across latent product segments.

Taken together, prior work supports three motivating ideas: (1) discount depth may influence customer perception, (2) review text contains measurable satisfaction information, and (3) buyer response to markdowns may differ across product contexts. Building on these observations, this project combines review text, numeric discount signals, and product metadata in a unified pattern-recognition pipeline for regression, classification, and clustering analysis.

Project novelty: Prior literature has not extensively examined explicit numeric discount depth as a continuous structured signal alongside review text in this specific predictive setting. Table 1 summarizes the analytical components incorporated here.

Analytical Dimension	Included in This Project	Why It Matters
Review Text Features	Yes (TF-IDF)	Captures customer language and product experience signals
Numeric Discount Signals	Yes (percentage, absolute, ratio)	Encodes explicit pricing context beyond text
Product Metadata	Yes (category, log price)	Controls for product-type and price-level differences
Ablation Comparison	Yes	Quantifies the contribution of each modality
Supervised Prediction	Yes	Evaluates rating and sentiment predictability
Unsupervised Segmentation	Yes	Examines whether discount–satisfaction patterns vary by segment

Table 1. Summary of analytical components incorporated in the proposed multimodal pattern-recognition pipeline.

3. Methodology

3.1 Dataset

We use the Amazon Sales Dataset from Kaggle (1,465 rows × 16 columns). After coercing non-numeric ratings and dropping nulls, 1,464 rows remain. Fields include *actual_price*, *discounted_price,*

discount_percentage, rating (2.0–5.0, σ=0.292), *review_content*, *review_title, category,* and *product_id*. Prices are in Indian Rupees (₹). The dataset spans Electronics, Home & Kitchen, Computers & Accessories, and Office Products.

3.2 Feature Engineering

We engineer four numeric discount features, each capturing a different pricing signal: **discount_percentage** (relative markdown), **abs_discount** (actual − discounted price; absolute savings matter more for cheap items), **discount_ratio** (discounted/actual; normalized 0–1), and **log_price** (log(1+actual_price); compresses skewed distribution). Review text is vectorized using TF-IDF with 20,000 features and English stop-word removal.

Sentiment label design. The sentiment label is derived from the star rating: rating ≥ 4.0 set as positive (1), otherwise negative (0), yielding a 76/24 class split. This avoids a critical flaw in the initial design: using VADER (a rule-based sentiment tool) on the review text and then predicting those labels from the same text is circular. The 4.0 threshold is natural given the left-skewed distribution (mean=4.1). The resulting 76/24 split is much healthier for classification than the 94/6 split VADER produced.

3.3 Pipeline Architecture

A scikit-learn ColumnTransformer applies TF-IDF to review_content, StandardScaler to the four numeric columns, and OneHotEncoder to category, all in parallel. The train-test split uses GroupShuffleSplit (80/20) grouped by product_id to prevent data leakage: all reviews of a given product appear exclusively in either train or test. Without this, the model could memorize product-specific patterns and produce inflated performance estimates.

3.4 Models

Baselines

- Ridge Regression (α=1.0) - L2 regularization prevents overfitting on 20k sparse TF-IDF features.
- Balanced Linear SVC (C=1) - maximizes the margin in high-dimensional space; balanced weights upweight minority-class errors.

Upgrades

- XGBRegressor (max_depth=3 to limit overfitting)
- HistGradientBoostingRegressor (early stopping),
- RandomForestClassifier,
- XGBClassifier.

Clustering

MiniBatch k-Means (k=8) on TF-IDF vectors reduced to 100 dimensions via TruncatedSVD. Cluster interpretation via lift-based term analysis.

3.5 Evaluation Metrics

Regression: **MAE** and **RMSE**.

Classification: **Macro-F1** (equal weight per class which prevents the model from coasting on the majority class) and **accuracy**.

Clustering: **silhouette score** and one-way **ANOVA** on cluster mean ratings. All metrics are compared against dummy baselines. Stability is verified via 5-fold grouped cross-validation.

4. Implementation Details

The project is implemented in Python 3.12 using Google Colab. Key libraries: scikit-learn (1.4+), xgboost (2.0.3), pandas, numpy, matplotlib, seaborn, scipy. The entire pipeline runs sequentially where one click can execute all cells with progress indicators. Estimated runtime: under 5 minutes on standard Colab CPU.

The ablation study uses a factory function (make_pipeline) that conditionally includes text and/or numeric blocks, ensuring identical preprocessing across text-only, discount-only, and combined variants. Category is retained in all variants as metadata, isolating the text vs. discount comparison. A custom ToDense transformer converts sparse TF-IDF matrices for tree-based models. Cluster interpretation uses lift defined as P(word|cluster)/P(word|not cluster) , rather than raw frequency, which would surface generic terms like "good" in every cluster.

Assumptions: (1) Star rating is a valid satisfaction proxy. (2) The 4.0 sentiment threshold is natural given mean=4.1. (3) TF-IDF captures sufficient text signal without word order; transformers would overfit on 1,464 rows. (4) k=8 balances granularity with interpretability (validated in Section 5.5).

Cluster selection - I evaluated k from 3 to 12 using elbow (inertia) and silhouette analysis. Although k=12 yielded the highest silhouette score, all silhouette scores are low (<0.07) due to overlapping review language across product types. However, ANOVA validates the choice post-hoc: for k=8, inter-cluster rating differences are highly significant ($p=1.51\times10^{-8}$). I selected k=8 as it produces clusters large enough for statistical analysis while remaining interpretable and acts as a practical balance between interpretability, cluster size, and downstream statistical comparison

5. Experimental Results

5.1 E1: Baseline Performance

Experiment	Metric	Value	vs. Dummy
E1-R: Ridge (combined)	MAE	0.157	−26%
E1-R: Ridge (combined)	RMSE	0.216	-
E1-R: Dummy (mean prediction)	MAE	0.212	-
E1-S: Balanced SVC	Macro-F1	0.716	+73%
E1-S: Balanced SVC	Accuracy	0.776	-
E1-S: Dummy (majority class)	Macro-F1	0.415	-

Table 2. Baseline results vs. dummy baselines.

Ridge achieves MAE=0.157, a 26% reduction over the dummy (0.212). Given σ=0.292, the average error is about half a standard deviation. The balanced SVC achieves Macro-F1=0.716, nearly doubling the dummy's 0.415. The confusion matrix shows 54% recall on negatives (45/83), confirming balanced weights address the 76/24 imbalance.

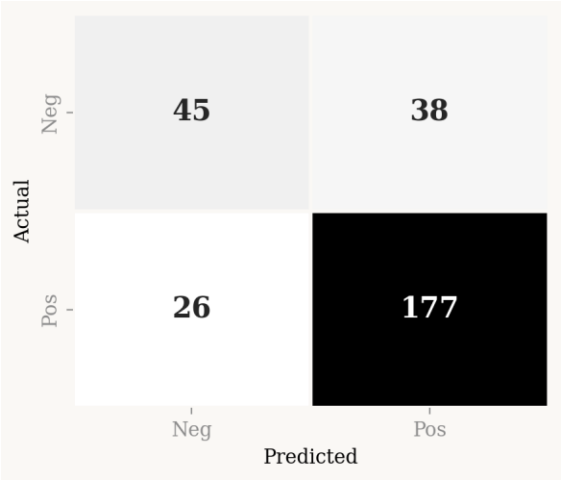

Figure 1. Confusion matrix for balanced SVC sentiment classifier.

Practical interpretation. An MAE of 0.157 on a 5-point scale means predictions are off by about 0.16 stars on average. For a merchandiser, this precision suffices to distinguish satisfactory (4.0+) from problematic (<4.0) segments. The classifier identifies over half of dissatisfied customers, while not perfect, is far better than ignoring the minority class.

5.2 E2: Ablation Study

Variant	Rating MAE ↓	Sentiment F1 ↑
Text-only	0.169	0.700
Discount-only	0.172	0.702
Combined	0.157	0.716

Table 3. Ablation: text-only vs. discount-only vs. combined.

Combined outperforms both single-modality variants on both tasks. Rating MAE improves 7% over text-only (0.157 vs. 0.169). Sentiment F1 rises from 0.700 to 0.716. This directly answers RQ1 and RQ2: discount features add predictive signal beyond text alone.

A striking finding: discount-only features perform comparably to text-only (MAE 0.172 vs. 0.169; F1 0.702 vs. 0.700). Pricing signals carry nearly as much information about satisfaction as the full review text. A merchandiser could roughly estimate satisfaction from pricing data alone, before reviews are written, enabling proactive pricing adjustments.

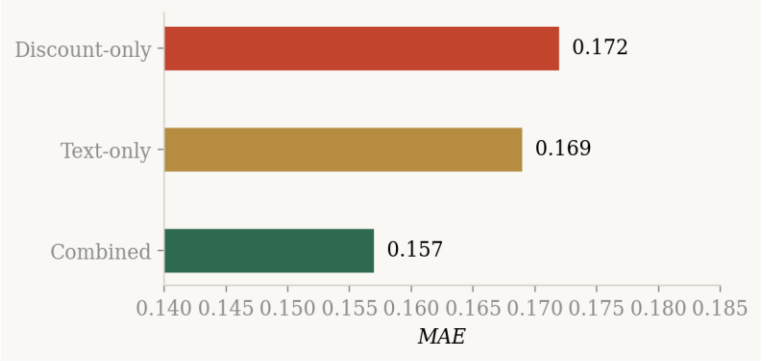

Figure 2. Rating regression ablation. Combined features yield lowest MAE.

5.3 E3: Model Upgrades

Model	Task	Metric	Value

Ridge (baseline)	Rating	MAE	0.157
XGBRegressor	Rating	MAE	0.167
HistGB	Rating	MAE	0.186
Balanced SVC (baseline)	Sentiment	F1	0.716
RandomForest	Sentiment	F1	0.549
XGBClassifier	Sentiment	F1	0.518

Table 4. Linear baselines vs. tree ensembles.

Trees underperform on both tasks due to: (1) small data (~1.4k rows) causing overfitting, (2) sparse 20k-dimensional TF-IDF features limiting split quality, and (3) narrow target range (σ=0.29) offering little non-linear signal. This is an informative methodological finding: for small multimodal datasets with sparse text features, regularised linear models generalise better than complex ensembles.

5.4 Cross-Validation

Five-fold grouped cross-validation yields mean MAE = 0.164 ± 0.010. The low variance confirms results are stable and not an artifact of a lucky split.

5.5 E4: Segment Discovery

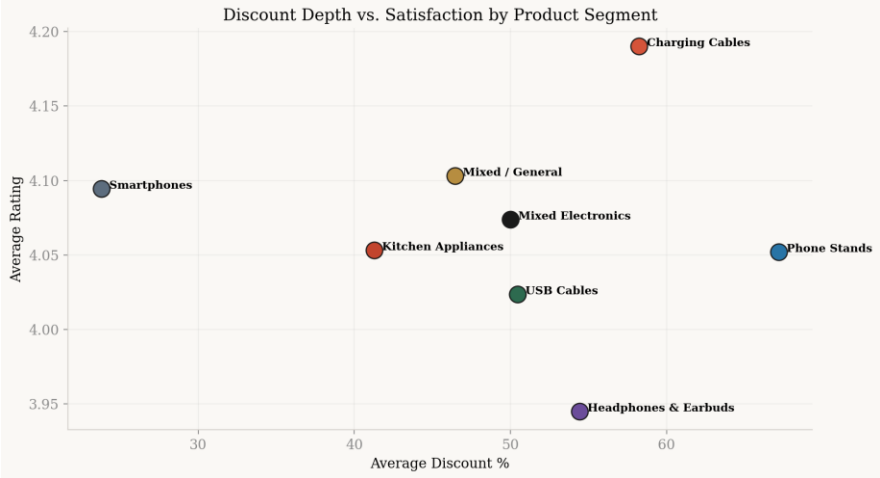

Figure 3. Discount depth vs. satisfaction by product segment. Labels derived from lift analysis and validated against product subcategories.

MiniBatch k-Means (k=8) on SVD-reduced TF-IDF vectors yields interpretable soft clusters that approximately align with distinct product types.. Notably, clusters discovered purely from review text align closely with actual product subcategories, validating that TF-IDF encodes product-type semantics. ANOVA yields F=7.24, $p=1.51\times10^{-8}$, confirming statistically significant inter-cluster rating differences.

C#	N	Rating	Disc %	Avg Price	Product Segment	Key Insight
0	105	4.05	41%	₹4,190	Kitchen Appliances	Moderate discounts, moderate satisfaction
1	826	4.10	46%	₹6,071	Mixed / General	Largest cluster; baseline behavior
2	17	4.02	50%	₹425	USB Cables (cheap)	Small cluster, average satisfaction
3	89	3.94	54%	₹2,655	Headphones & Earbuds	Lowest rating (signaling deal-bait pattern)
4	23	4.05	67%	₹1,644	Phone Stands & Mounts	Deepest discounts, average rating

5	194	4.19	58%	₹1,056	Charging Cables (branded)	Highest rating (signaling discounts work here)
6	72	4.09	24%	₹19,837	Smartphones	Lowest discounts, quality-driven buyers
7	138	4.07	50%	₹4,389	Mixed Electronics	Image-heavy reviews, moderate

Table 5. All 8 product segments with ratings, discounts, and key insights.

Three patterns answer RQ3. First, Cluster 6 (Smartphones) has the highest price (₹19,837) and lowest discounts (24%), yet customers are satisfied (4.09). These quality-driven buyers need no markdowns. **Second,** Cluster 3 (Headphones & Earbuds) shows the "deal-bait" pattern: lowest ratings (3.94) despite 54% discounts. Deep markdowns cannot fix quality issues, validating Lichtenstein and Burton's quality-signaling hypothesis [2]. **Third,** Cluster 5 (Charging Cables) achieves the highest ratings (4.19) at 58% discounts. This segment genuinely responds well to markdowns - customers feel they received both a deal and a quality product.

Figure 4. Cluster feature heatmap across all 8 segments.

Practical implication: a merchandiser should not apply a uniform discount strategy. Smartphone customers are quality-driven and need no deep discounts. Cable customers respond well to markdowns. Audio product ratings are low regardless of discount depth implying the problem is the product, not the pricing.

6. Conclusion and Future Research Directions

This project demonstrates that explicit numeric discount features add measurable predictive signal to both rating regression and sentiment classification on Amazon listings. The ablation study provides core evidence: combined features outperform text-only and discount-only variants on both tasks. Discount-only features carry nearly as much information as full review text which is a finding with immediate practical value for pricing teams.

Linear models outperform tree ensembles on this small, sparse dataset which serves as a useful methodological insight. Unsupervised clustering reveals product-type segments with significantly different discount–satisfaction relationships (ANOVA $p<10^{-8}$). Three distinct patterns emerge: quality-driven segments (Smartphones) needing no discounts, deal-responsive segments (Charging Cables) where discounts work, and deal-bait segments (Headphones) where deep discounts co-occur with low ratings.

Limitations. (1) Only 1,464 rows from Amazon India. (2) No temporal features for seasonal effects. (3) Low silhouette score (0.022) indicates soft cluster boundaries. (4) No user-level modeling.

Future work. (1) Scale to 100k+ rows to enable BERT embeddings. (2) SHAP analysis on Ridge for interpretability. (3) Time-series analysis of seasonal discount effects. (4) Cross-marketplace comparison (US vs. India). (5) Causal inference to distinguish correlation from causation in discount–satisfaction.

Project Link

https://github.com/SidraFarooqui/Pattern_Recognition_Project_Amazon-Discounts-vs.-Satisfaction.git

References

[1] Thaler, R. (1985). Mental accounting and consumer choice. Marketing Science, 4(3), 199–214.

[2] Lichtenstein, D. & Burton, S. (1989). The relationship between perceived and objective price-quality. J. Marketing Research, 26(4), 429–443.

[3] Grewal, D., et al. (2017). The future of retailing. J. Retailing, 93(1), 1–6.

[4] McAuley, J. & Leskovec, J. (2013). Hidden factors and hidden topics. Proc. RecSys, 165–172.

[5] Pang, B. & Lee, L. (2008). Opinion mining and sentiment analysis. Found. & Trends in IR, 2(1–2), 1–135.

[6] Simester, D., et al. (2019). Targeting prospective customers. Intl. J. Research in Marketing, 36(4), 637–651.